The Tech Interactive | Visitor Companion

Project brief • Responsive web app • Proposed student team: 5 developers, 3 designers, 1 UX researcher

The idea

Build a playful companion to a visit at The Tech Interactive. Guests can discover exhibits, choose what to explore next, and collect accessories for the Tech mascot as they go. The app should make the museum easier to navigate on the day of a visit—and give families a reason to return for new exhibits, seasonal events, and collectibles.

What success looks like

A guest can open the app on a phone, tablet, or laptop; find an exhibit that interests them; mark an exhibit as visited; and see their mascot change. On a later visit, their progress is still there, with new things to explore. The experience should be fun without making a login or download mandatory.

Core experience

Explore the museum. Browse exhibit cards with a short description, topic, age guidance when available, location, and a photo. Filter by interest, time available, or exhibit type.

Collect and customize. Visitors earn a badge or mascot accessory for completing an exhibit check-in. A simple collection screen shows what they have unlocked and what is still left to discover.

Find your next stop. A recommendation feature suggests 2–3 relevant exhibits based on interests, exhibits already explored, and optional visit time. Explain each suggestion in plain language.

Come back for something new. Feature rotating exhibits and seasonal experiences such as Tech or Treat, with limited-time collections and a “what’s new since your last visit” section.

Save the memories. An optional visit scrapbook keeps completed exhibits, badges, favorite moments, and a small photo gallery using images the guest chooses to add.

Mascot direction



The supplied mascot concepts are a starting point for a small, reusable accessory system—not a requirement to animate every character.

Scope & delivery approach

A practical first release that can be expanded after testing with visitors.

Recommended additions

Choose-your-own adventure: visitors pick a theme such as “build,” “nature,” or “space,” and get a short suggested trail across exhibits. **Mini discovery prompts:** one quick observation or question at each stop, rather than a full game. **Return-visit highlights:** a personalized list of new exhibits, new accessories, and seasonal activities since the last visit. These are lightweight ways to encourage exploration without relying on leaderboards or constant notifications.

What we will build first

Release 1: responsive exhibit directory, exhibit detail pages, mascot profile, collectible badge/accessory rules, visit history, simple recommendation logic, seasonal content section, and basic admin-friendly content format. **If time allows:** scrapbook gallery, themed trails, optional account sync, and richer recommendation ranking. No indoor positioning, ticketing integration, or museum system integration is assumed for the first release.

How it works technically

A mobile-first progressive web app (PWA) works in modern iOS and Android browsers and on iPad and laptop screens. A React/Next.js front end can use an accessible component library and lightweight animations. A small API and database store exhibit metadata, seasonal collections, and—when guests opt in—visit progress. For the pilot, visitors can check in by selecting “I explored this”; exhibit-specific QR codes can be added later if practical. Recommendations start with transparent, rules-based matching, so no training data or AI service is required. Content can be maintained through a simple structured sheet or lightweight admin view.

Team plan

5 developers: 2 on responsive exhibit discovery and UI; 2 on progress, collectibles, data and API; 1 on recommendations, seasonal content and integration/testing.

3 designers: 1 on visual system and mascot accessory treatments; 1 on exhibit discovery and responsive layouts; 1 on progress, scrapbook and interaction states.

1 UX researcher: Lead short visitor/staff interviews, test key flows on phones and tablets, and synthesize feedback into prioritized changes.

Suggested 10-week timeline

Weeks 1–2: confirm exhibit content and visitor journeys; wireframes and technical setup. **Weeks 3–4:** design system, exhibit pages, and data model. **Weeks 5–7:** check-ins, mascot collection, recommendations, and seasonal content. **Weeks 8–9:** device testing, accessibility, user testing, and refinements. **Week 10:** pilot handoff, documentation, and demo.

What we need from The Tech Interactive

A list of current exhibits (names, short descriptions, categories, and locations), any existing photos and approved mascot artwork, upcoming seasonal exhibit dates, and one point of contact for occasional content checks. A short kickoff and a few asynchronous reviews should be enough; we can prototype with sample exhibit data while assets are gathered. Any public launch or use of official branding would be reviewed before release.

Design priorities

Bright, friendly, and consistent with the existing mascot; readable for families; large touch targets; clear progress feedback; accessible contrast; reduced-motion support; and a layout that feels natural on both phones and larger screens.